

The Journal of International Trade & Economic Development

An International and Comparative Review

ISSN: 0963-8199 (Print) 1469-9559 (Online) Journal homepage: www.tandfonline.com/journals/rjte20

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To cite this article: Su Pan & Zhaohui Chong (2023) Effects of FDI on trade among countries along the Belt and Road: A network perspective, The Journal of International Trade & Economic Development, 32:1, 84-103, DOI: [10.1080/09638199.2022.2078865](https://doi.org/10.1080/09638199.2022.2078865)

To link to this article: <https://doi.org/10.1080/09638199.2022.2078865>



Published online: 06 Jun 2022.



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Effects of FDI on trade among countries along the Belt and Road: A network perspective

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ABSTRACT

From the network perspective, this paper explores the effect of FDI on trade and analyzes the structural effect on different sectors classified by technology levels among the Belt and Road (B&R) countries. Employing social network analysis methods, we depict the evolution of FDI networks and trade networks of B&R countries and find that both two networks are disassortative and small-world networks as well as the density and number of edges increase. Through the results of ERGM, we get three findings about the effect of FDI on trade. (1) FDI has the positive effect on trade among B&R countries from the network perspective; (2) the positive effect of FDI on trade has improved since the Belt and Road Initiative is proposed; (3) FDI promotes the exports of low technological sector, medium technological sector, and high technological sector, but the effect on medium technological sector is stronger than the other two sectors.

KEYWORDS FDI; trade; networks; Exponential Random Graph Model (ERGM); Belt and Road Initiative

JEL CLASSIFICATIONS F21, F40, R11

ARTICLE HISTORY Received 6 September 2021; Accepted 12 May 2022

1. Introduction

Under the influence of global COVID-19 pandemic, the ongoing processes of global fragmentation would be deepened and accelerated (Abdal and Ferreira 2021). The globalization has shown new characteristics and has gradually changed from the multilateral economic and trade cooperation to the regional economic cooperation. The Belt and Road Initiative, first proposed by President Xi Jinping of China in 2013, aims to improve trade and investment links among the B&R countries.¹ Since 2014, China and other countries have established new financial entities providing investment for the B&R countries, such as the Silk Road Fund financed from China's foreign reserves, the New Development Bank financed by the BRICS countries and the Asian Infrastructure Investment Bank financed from the B&R founding countries (Aoyama 2016). Exports from these 45 economies² account for about 65.36% of total world exports and about 60.38% of world imports. Can FDI foster trade among the B&R countries? Is there heterogeneous

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effect on trade of FDI among the B&R countries? These are issues worthy of attention which can show how B&R initiative would be positive for the region.

The relationship between FDI and trade has been concerned since Mundell's (1957) seminal work. In the traditional Heckscher-Ohlin trade theory, FDI has a substitutive effect on trade. The new trade theory reveals that FDI can have both a substitutive and a complementary effect on trade depending on the circumstances (Dunning 1980; Markusen 1997; Helpman, Melitz, and Yeaple 2004). Meanwhile, there is also no agreement in the empirical analysis. The results vary across countries (Jawaid et al. 2016; Ahmad, Draz, and Yang 2018), regions (Fetai and Morina 2018; Sohail, Zatullah, and Li 2021), industries (Hejazi and Safarian 2001; Chen, Hsu, and Wang 2012; Bouras and Raggad 2015), firms (Head and Ries 2001; Lim and Moon 2001), and products (Lipsey and Weiss 1981; Nishitaten 2013) over time. In recent years, scholars show a growing interest in employing Social Network Analysis to analyze the international FDI network (IFN) and the international trade network (ITN). The studies of IFN and ITN consider countries around the world as nodes and their possible financial or trade interaction as links in a graph, mainly focusing on their topological properties (density and average degree, centralization by outdegree and indegree, average path length, assortativity, small world property) (Serrano and Boguna 2003; Garlaschelli and Loffredo 2005; Bolívar, Casanueva, and Castro 2019; Arif et al. 2021) and their evolution over time (Bhattacharya et al. 2008; Fagiolo, Reyes, and Schiavo 2009; Zhang et al. 2016; Li, Liao, and Sun 2018). However, few studies analyze the effect of FDI on trade from the network perspective. To fill this gap, we attempt to investigate the effect of FDI on trade from a network perspective, which can provide new insight on FDI-trade nexus.

This paper provides three contributions as follows:

- (1) The evolution pattern and structural characteristic of FDI and trade networks are detected from a dynamic network perspective. The studies focusing on the bilateral perspective ignore the multilateral dependencies of trade or financial relationships among countries (Aizenman and Noy 2006; Liu et al. 2016). The network approach gives a unified view of the system characteristics, which allows us to identify the whole structure of interactions among countries while the standard statistics are only able to recover the import/export intensity between two countries (Schiavo, Reyes, and Fagiolo 2010; De Benedictis and Tajoli 2011; Liu et al. 2018). Considering that FDI can foster technology transfer, we estimate the different effects of FDI on trade among different technological sectors by Lall's (2000) classification, which are also presented by trade networks of high technology sector, medium technology sector, and low technology sector. Our results reflect the obvious sectoral disparity in the B&R countries, that is, the medium technology sector beneficiary of FDI after the Belt and Road Initiative is assigned.
- (2) Exponential Random Graph Model (ERGM) is employed to analyze the effect of FDI on trade. ERGM are statistical models in the Social Network Analysis which can be seen as similar to regression models, but they can avoid problems of standard correlation in linear regression analysis by taking the structural dependency into account. One of the basic assumptions of OLS linear regression analysis is independence of observations, and this can lead to biased estimates of the gravity equation (Broekel et al. 2014). The network method has a high degree of accuracy compared with the structural gravity model in terms of root mean square error (Dumor et al. 2021). Hence, we apply the ERGM which is a statistical approach to model networks

by supposing the interdependence between network links and is also appropriate for dyadic data.

- (3) The third contribution of the paper is that we focus on the dynamic relationship between FDI and trade. As many researches point out, the relationship between FDI and trade responds to changing conditions in a dynamic way (Büthe and Milner 2008; Forte and Silva 2017). As the time span of our research is from 2009 to 2015, we can compare the change of effect before and after the Belt and Road Initiative is proposed. Hence, how FDI influences trade before and after the Belt and Road Initiative is analyzed.

The paper is organized as follows: Section 2 provides the literature review on the effects of FDI on trade and addresses some hypotheses. Section 3 reports and compares the evolution characteristics of the IFN and ITN after a presentation of the data. Then, we identify the effects of FDI on trade focusing on the sample of 45 B&R countries by using the ERGM and discuss the results in Section 4. Section 5 draws conclusions and offers policy implications.

2. Literature review and hypotheses

2.1. *Studies on the relationship between FDI and trade*

Some researchers analyzed the relationship between FDI and trade from the bilateral perspective. In the case of Sweden, Austria, Spanish, Pakistan, and China, a complementary relationship is confirmed between FDI and exports (Bajo-Rubio and Montero-Muñoz 2001; Liu et al. 2016; Jawaid et al. 2016). The studies of Hejazi and Safarian (2001) and Türkcan (2007) confirm the complementary relationship between FDI and exports using micro data and other methodologies. However, several studies find evidence of substitute relationship between FDI and exports. Blonigen (2001) finds evidence of substitution by using the product-level data on a set of Japanese-produced final consumer goods. Similarly, Mitze, Alecke, and Untiedt (2010) find the substitutive FDI-trade relationship between German with the EU-27. Overall, there is no agreement related to the relationship between FDI and exports.

Although the bilateral analysis is adopted by many researchers, a multilateral approach is needed for a complete analysis of relationship between FDI and trade (Lipseý, Ramstetter, and Blomström 2000). From the multilateral perspective of countries, the relationship between FDI and trade is mostly complementary. Most of the scholars testify the effect of FDI on trade among European Union member states (Dritsaki and Dritsaki 2012; Pourshahabi, Salimi Soderjani, and Mahmoudinia 2013). As stated by Ahmad, Draz, and Yang (2018), the complementary effects of Outward Foreign Direct Investment (OFDI) on exports outweigh the substitution effects for four ASEAN countries during 1981–2013. Ibrahim (2020) examines the effects of trade and financial integration in 32 African countries and finds that trade and financial integration effects complement each other from 1985 to 2015. From the view of the global trade and FDI network comprising 75 countries from different regions in the world, Baronchelli and Uberti (2018) find that there is a positive association between world trade network and world investment network, which means that FDI and exports could be complements, rather than substitute.

According to the official data in the Ministry of Commerce of the People's Republic of China, Chinese cumulative investments in the B&R countries amount to USD 177.9 billion, with the growth rate of 18.3% in 2020. The increasing investments are likely to promote the trade in the B&R countries in several ways: the upgrading of critical infrastructure will fill gaps in trade and production networks, which helps reduce trade costs and increase trade flows in the B&R region, particularly the land-locked countries (Enderwick 2018). FDI in B&R countries can contribute to the export of these countries by increasing supply and output capacity (Fetai and Morina 2018) and reducing the trade cost by upgrading transport infrastructure (Baniya, Rocha, and Ruta 2020). Baniya, Rocha, and Ruta (2020) find that the Belt and Road Initiative increases trade flows among countries by up to 4.1% because of the upgrading transport infrastructure. Besides that, industrial relocations from China to the B&R countries will encourage local production and help countries establish an investment-and-export-led development model (Li, Liao, and Sun 2018).

Based on that, we propose our first hypothesis:

Hypothesis 1. From the perspective of networks, FDI would stimulate goods trade among the B&R countries.

2.2. Studies on the dynamic nature of the relationship between FDI and trade

As many studies have shown, the relationship between FDI and trade is not static but varies over time. Büthe and Milner (2008) find that an initial complementary relationship between FDI and export is turned into a substitute relationship when studying the US firms, since internationalization advances to the high degree. According to the study of 11 OECD countries since 1971 by Pain and Wakelin (1998), FDI is taken as a deliberate measure of serving foreign markets from foreign production when trade and capital market barriers have been lowered and hence the negative relationship between FDI and export has strengthened over time. Jawaid et al. (2016) find a changing relationship between FDI and exports from 1974 to 2012 in Pakistan. Furthermore, Lim and Moon (2001) take the industry's life cycle into consideration and conclude that the relationship between FDI and net-export is only positive and significant for the FDI that lasts for less than 5 years.

The Belt and Road Initiative significantly reduces shipment times and trade costs by 3.2% and 2.8% for the B&R countries, respectively (De Soyres et al. 2019). Stefano et al. (2021) find that the intensity of trade between B&R countries will increase when they experience improvements in their infrastructure systems. Consequently, we propose the second hypothesis:

Hypothesis 2. The Belt and Road Initiative will further promote the impact of FDI on trade.

2.3. Studies on the structural effects of FDI on trade

Multinational companies have a better knowledge about international foreign markets, advanced technology, and better supply chain management. FDI will transfer its intangible assets such as information, knowledge, and technology to affiliates, which may diffuse to local companies and improve their efficiency of production, especially to the export-oriented firms (Sinani and Meyer 2004; Jithin and Babu 2020). Eventually, technology transfer of FDI will be reflected in the export of different technology sectors.

Chen, Hsu, and Wang (2012) dividing their sample into the ‘traditional sectors’ and ‘modern sectors’ based on the median of R&D expenditure per employee, find that the positive relationship only in China for the ‘modern sectors’ and the export is stronger in ‘traditional sectors’ than in ‘modern sectors’. Lipsey and Weiss (1981, 1984) find that the positive relationships are for seven sectors (Textiles and apparel, Printing and publishing, Agricultural chemicals, Office and computing machinery, Electronic components, Instruments, and Other manufacturing) and the negative relationships are for four sectors (Other food products, Drugs, Primary nonferrous metals, and Lumber, wood, furniture, and fixtures). Türkcan (2007) classifies 2156 items as final goods and 1079 items as intermediate goods from the 5-digit level of Standard International Trade Classification (SITC) with the results of the complementary relationship between FDI and intermediate goods and the weak evidence of substitution effects between FDI and final goods; Sgrignoli et al. (2017) use the network measures and find that FDI has a positive impact on trade both directly and indirectly, while the indirect effects are more pronounced for manufacturing sectors than for primary sectors (such as oil extraction and agriculture).

As to the B&R countries, there are strong sectoral disparities in the exports and imports. According to the findings of Cui and Song (2018), the Belt and Road Initiative can significantly improve economic and trade cooperation between China and surrounding countries. The exports of China’s textile, ferrous, transport equipment, and machine equipment industries have increased, while the exports of China’s electronic equipment and other manufacturing industries have decreased.

All the above leads us to propose our third hypothesis:

Hypothesis 3. Considering the technology spillover, the effect of FDI on trade would have sectoral heterogeneity among the B&R countries.

3. Description of the IFN and ITN networks

3.1. Data

FDI data employed in the paper are taken from the IMF Coordinated Direct Investment Survey (CDIS). The survey is conducted annually starting with data for end-2009 and gives information on the amount of ‘inward’ or ‘outward’ direct investment issued by all partner countries held by each of the 93 reporting countries. We focus attention on data covering 45 countries (or regions) along the B&R for the 2009–2015 period. The generic element (i, j) of each matrix is the flow of FDI from country i to country j in a given year. Thus, we construct a multi-valued network. By setting different thresholds of the network linkages, we find that the number of linkages with higher amount is much less than the number of linkages with lower amount, which conforms to the principle of the 80/20 theorem. However, this imbalance is alleviated over time, which can also be seen in Figure 5.

As to trade, the data are assembled from the UN Comtrade database, which covers the same countries and the same period as the CDIS and allows us to compare the change of IFN and ITN over time. The generic element (i, j) of each matrix is represented by the exports from country i to country j . Using Lall’s (2000) classification by technology level, we divide the sample into ‘high technology sector’, ‘medium technology sector’ and ‘low technology sector’. High technology sector contains the products having advanced and

Table 1. Small world properties of the IFN and ITN.

		2009	2010	2011	2012	2013	2014	2015
IFN	<i>L</i>	1.459	1.392	1.377	1.370	1.325	1.331	1.311
	<i>CC</i>	0.711	0.754	0.761	0.764	0.792	0.789	0.794
	<i>SW quotient</i>	2.210	2.166	2.042	1.977	2.053	1.982	1.975
ITN	<i>L</i>	1.142	1.123	1.128	1.138	1.116	1.127	1.115
	<i>CC</i>	0.909	0.918	0.916	0.904	0.921	0.916	0.921
	<i>SW quotient</i>	1.423	1.403	1.381	1.516	1.380	1.416	1.364
High technology sector	<i>L</i>	1.273	1.265	1.255	1.266	1.254	1.273	1.254
	<i>CC</i>	0.820	0.822	0.828	0.820	0.824	0.816	0.829
	<i>SW quotient</i>	1.681	1.684	1.624	1.812	1.734	1.734	1.726
Medium technology sector	<i>L</i>	1.204	1.195	1.182	1.192	1.174	1.178	1.170
	<i>CC</i>	0.865	0.874	0.886	0.870	0.890	0.885	0.890
	<i>SW quotient</i>	1.476	1.480	1.459	1.617	1.479	1.498	1.488
Low technology sector	<i>L</i>	1.181	1.168	1.166	1.177	1.154	1.159	1.146
	<i>CC</i>	0.886	0.890	0.896	0.885	0.899	0.895	0.904
	<i>SW quotient</i>	1.481	1.468	1.438	1.577	1.451	1.471	1.442

Source: Authors.

fast-changing technologies; medium technology sector is comprised by the intermediate products and capital goods with scale-intensive technologies; low technology sector includes the products having stable and well-diffused technologies and undifferentiated characteristics. The technological classification is given in Table 1 in the appendix. By classifying the sample in this way, the impact of FDI on trade depending on industry characteristics can be explored.

In this analysis, we apply social network analysis method to depict the networks of FDI and exports. For the whole network, we compute the indices describing networks as a whole: *density*, *edge number*, *edge strength*, *assortativity*, *average path length*, *overall clustering coefficient*. For nodes, we compute *indegree* and *outdegree centrality*. Using the binary data, we calculate the most common whole network and node indices for all 30 matrices (6 networks for FDI, 6 networks for exports, 6 networks for high technology sector, 6 networks for medium technology sector and 6 networks for low technology sector from the year 2009 to 2015).

3.2. Characteristics of the IFN and ITN from the view of the whole network

3.2.1. Density and strength

Figures 1 and 2 present the edge strength and the densities of IFN and ITN from 2009 to 2015. The number of nodes between 2009 and 2015 stays at 45 consistently. First, the IFN becomes denser in this period, as the number of edges increases. The number of existing links grows from 715 in 2009 to 933 in 2015 and the density of the network increased from 0.361 to 0.471. 31 pairs of countries on average establish financial relations each year. While the edge strength temporarily decreases in 2014, it recovers quickly and increase to 2513.1 million US dollars in 2015. Generally, the financial relations and the volume of direct investment flows among the 45 countries are both increased from 2009 to 2015. As Arif et al. (2021) observe that the number of the nations in the world with FDI relationships is increasing from 2001 to 2012 and the nations in the IFN are getting closer and closer.

Second, the density of ITN has declined in 2012 and 2014 and the number of edges decreases to 1366 and 1445 respectively, while the edge strength increases to 772.5 and

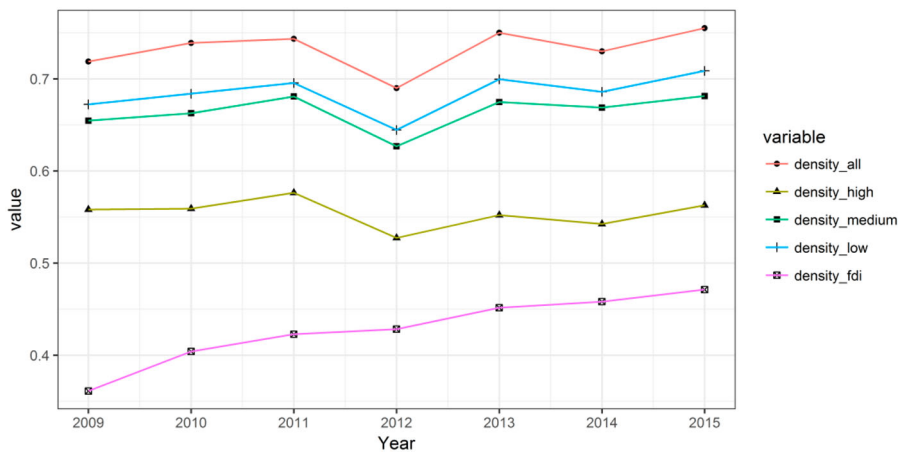


Figure 1. Comparison of densities between the IFN and ITN. Source: Authors.

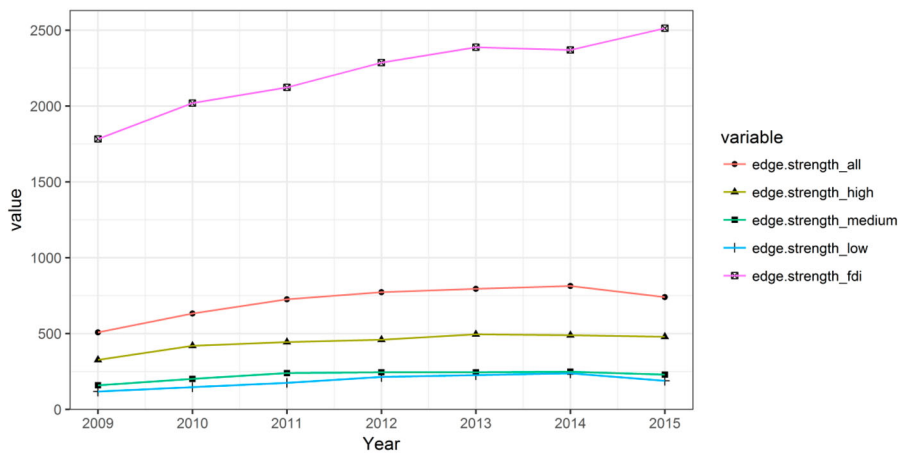


Figure 2. Comparison of edge strength between the IFN and ITN. Source: Authors.

813.5 million US dollars. Although the trade relations decline in the two years, the volume of trade flows is increased. This means that trade networks are more polarized with fewer but stronger relationships. Of the three sectors, low technology sector has the higher densities, ranging between 0.644 and 0.709. Medium technology sector displays the densities from 0.627 to 0.681, while the densities of high technology sector is just above 0.500. Therefore, the trade of low technology sector is substantially more widespread among the countries in the sample.

3.2.2. Assortativity

In social networks, evidence suggests that similar nodes connect with similar nodes. In the IFN and ITN, we call a network ‘assortative’ if the well-connected countries tend to link with other well-connected countries (i.e. similar countries) and we call a network ‘disassortative’ if the well-connected countries that have high centrality scores

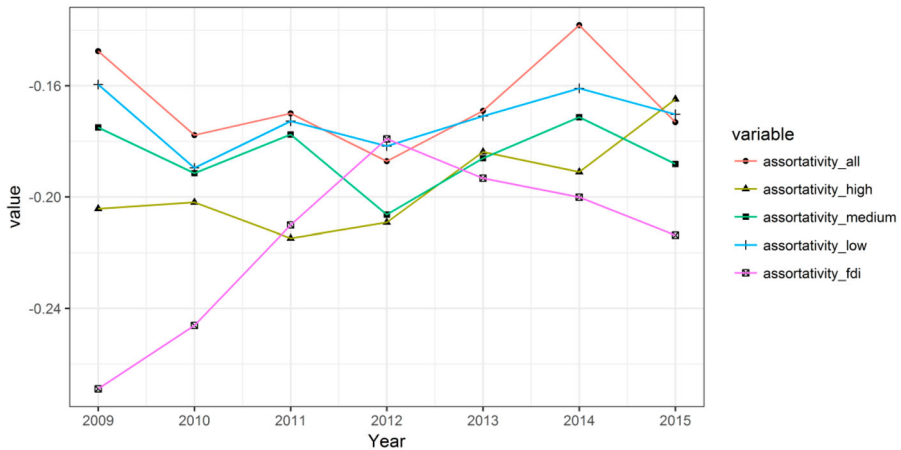


Figure 3. Comparison of assortativity between the IFN and ITN. Source: Authors.

associate more with poorly connected countries that have low centrality scores in the network. Assortativity conveys the information on the finer structure of the network, that is the tendency for nodes to be connected to other nodes that are similar or different to themselves in some way (Newman 2002).

Following Newman (2002), the assortativity coefficients are computed to verify if the IFN and ITN are assortative or not. The results in Figure 3 remain negative, which confirms that the IFN and ITN are disassortative networks. The results are in line with the related literature, as Schiavo, Reyes, and Fagiolo (2010) and Baronchelli and Uberti (2018) also find that both world investment network and world trade network are disassortative. Also, there is a stronger disassortative pattern in the high technology sector than the medium technology sector and the low technology sector and their assortative coefficients are on average -0.196 , -0.185 , and -0.172 , respectively.

3.2.3. Small world property

Due to that international trade is a mutually beneficial or cooperative activity, the distribution of trade relations is not either completely regular or completely random but shows ‘small-world’ property (Yu et al. 2015). Following Watts and Strogatz (1998), we use the average path length (L) and clustering coefficient (CC) to verify if the IFN and ITN have small-world properties or not. Specifically, L measures the separation between two vertices in the network, while CC measures the cliquishness of a typical neighborhood. In a small-world network, L is almost as small as in random network (i.e. $L/L_{random} \approx 1$) and CC is much higher than in random network (i.e. $C/C_{random} \geq 1$). The small-world quotient ($SWquotient$), represented by $(C/C_{random})/(L/L_{random})$, needs to be greater than 1. L_{random} is the characteristic path length, and C_{random} is the clustering coefficient of random networks with the same nodes as real networks. Comparing with the random networks, we find that the IFN and ITN are not random networks, but similar to small-world networks. The small-world quotients in both IFN and ITN are greater than 1, which means the reachability and connectivity of these networks are relatively high. Apparently, the small-world quotients of IFN are bigger than ITN, which indicates that investments are more concentrated in a few countries among the 45 B&R countries.

Besides that, we can also find that high technology exports are more concentrated than medium and low technology exports.

3.3. Centrality of the IFN and ITN

In our analysis, there are 45 B&R countries (or regions), including 6 ASEAN countries, 7 countries in West Asia, 6 countries in South Asia, 2 countries in Central Asia, 6 countries in Commonwealth of Independent States, 14 countries in Central and Eastern Europe, Mongolia, China, China: Hong Kong SAR and China: Macao SAR. Following Freeman (1978) and Deandrade and Rêgo (2018), we calculate the degree centrality which is one of the most common indicators to characterize the role and position of each node country in the networks. In a directed network, degree centrality can be distinguished as out-degree and in-degree centrality. In the IFN, out-degree centrality quantifies the tendency of countries to make investments to other countries and in-degree centrality quantifies the tendency of countries to receive investments from the other countries. In the ITN, out-degree centrality quantifies the tendency of countries to export to other countries and in-degree centrality quantifies the tendency of countries to import from the other countries.

According to the results of out-degree and in-degree centrality, we classify the countries into four groups. Group 1 contains the countries having the most activities in FDI and trade, which means that they have both high out-degree and in-degree centrality and have more influence than other countries in the IFN and ITN. Group 2 includes the out-flowing countries whose out-degree centrality is higher than in-degree centrality and Group 3 includes the in-flowing countries whose in-degree centrality is higher than out-degree centrality. Group 4 contains the countries having the minimum activities in FDI and trade, which means that they have the least financial and trade relationships with other countries in the IFN or ITN. Concrete groups of countries are given in Table A3 in the appendix.

Besides that, we also visualize the IFN and ITN of 45 B&R countries in 2009 and 2015, respectively. The size of circular icons in Figure 4 and Figure 5 indicates the different situational status of each country. The larger the circular icons are, the more important position the country has in the IFN or ITN. We can find that China occupies an important position in both IFN and ITN, with the high out-degree and in-degree centrality. China is the most important country with the most out-flowing and in-flowing investment in 2009 and 2015. Most countries of the Commonwealth of Independent States, such as Russia, Azerbaijan, Belarus, and Ukraine, have more in-flowing in FDI in 2009 and 2015 than out-flowing. Among them, only Russia has more imports than exports in both 2009 and 2015. For IFN, many countries in Asia have more out-flowing than in-flowing in FDI in 2009 and 2015, including Singapore, Turkey, India, Israel, and Malaysia. These countries are also important in the ITN, both in importing and exporting in 2009 and 2015.

4. Empirical analysis on the effects of FDI on trade

In the preceding section, we have explored the topological properties of IFN and ITN. In this section, we focus on the effects of FDI on trade by employing ERGM approach.

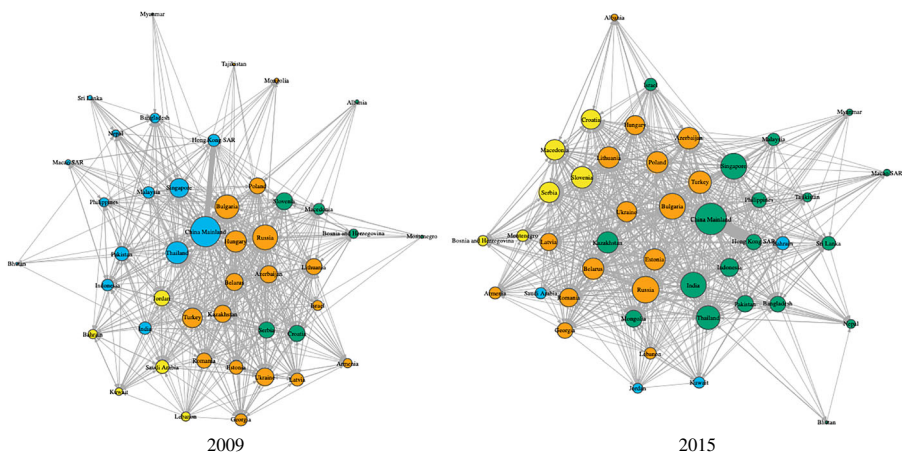


Figure 4. Comparison of IFN between 2009 and 2015. Source: Authors.
 Note: Colors of Node are different in different years. The nodes with the same color in a single year belong to the same cluster.

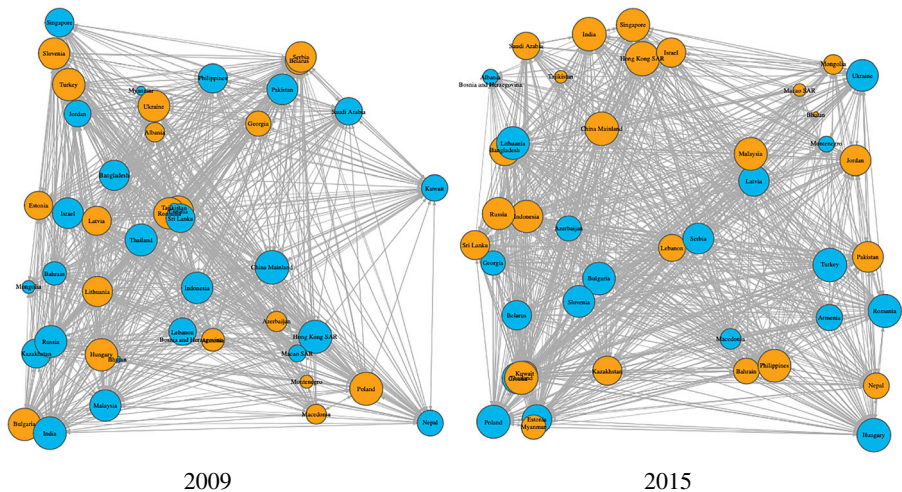


Figure 5. Comparison of ITN between 2009 and 2015. Source: Authors.
 Note: Colors of Node are different in different years. The nodes with the same color in a single year belong to the same cluster.

4.1. ERGM

A fundamental concept underpinning the ERGM is the dependency between network ties (Lusher, Koskinen, and Robins 2013). Considering that the assumption of independency in traditional statistical models is unlikely to hold (Lall 2000; Robins et al. 2007), we cannot use the OLS to analyze the impact of FDI on trade. However, ERGM whose assumption is dependency is suitable for the analysis. Therefore, we use ERGM to explore the effect of FDI on trade by analyzing the relationship between IFN and ITN.

Supposing that there are n nodes in a network which is represented by $V_i = [v_i] (i = 1, 2, \dots, n)$ and all possible relationships between nodes in the random network which are represented by $J = \{(i, j) : i, j \in V, i \neq j\}$. $G = (V, E)$ is an actual network, where E is a subset of J and represents the edges of the actual network. We employ a random variable Y to represent the elements of J , i.e. $y_{i,j} = 1$ if $(i, j) \in E$ and $y_{i,j} = 0$ otherwise. $y = [y_{i,j}]$ represents the adjacency matrix of the stochastic network and all adjacency matrices form the feasible set Y of the network adjacency matrix. Based on that, we use $\Pr(Y = y|\theta)$ to estimate the probability that y exists in Y under the condition θ . The general form of ERGM is as follows:

$$\Pr(Y = y|\theta) = \left(\frac{1}{k}\right) \exp \left\{ \sum_H \theta_H^T g_H(y) \right\} \quad (1)$$

where $k(\theta)$ is a normalizing constant and H represents all the possible factors influencing the formation of network. According to Lusher, Koskinen, and Robins (2013), the factors include endogenous structural factors, individual attributes, and other exogenous factors related to the network generally. The above formula can be specified as follows:

$$\Pr(Y = y|\theta) = \left(\frac{1}{k}\right) \exp \{ \theta_\alpha^T g_\alpha(y) + \theta_\beta^T g_\beta(y, x) + \theta_\gamma^T g_\gamma(y, \bar{g}) \} \quad (2)$$

where α indicates the importance of endogenous structural factors to the formation of network, β indicates the importance of node attributes and γ indicates the importance of other exogenous network factors.

We use the Monte Carlo–Markov maximum likelihood estimation method (MCMC MLE) to estimate the variable parameters which makes the parameters tend to be stable through continuous simulation and modification of the model.

4.2. ERGM variables

Combining with the above hypotheses, the ERGM variables include two endogenous structural factors, two individual attributes, and three exogenous factors.

Endogenous structural factors. Edges and reciprocity are included in ERGM as endogenous structural factors. Edges are the basic propensity for tie formation and capture the overall likelihood to establish trade relationships. The associated parameter is similar to intercepts in conventional regression analyses. The significant and negative value indicates that the formation of individual linkages is not completely random. Reciprocity captures the tendency of links to be mutual. The significant and positive value of the parameter indicates that if country A is exporting to country B, then there is always a higher probability for country B to export to country A to make a two-way reciprocal interaction.

Node attributes. The economic attributes of countries in networks can be very important to the formation of network ties (Lusher, Koskinen, and Robins 2013). We choose two variables of 45 countries in the sample as individual attributes: population and human capital, represented by *POP* and *HC*. The population and human capital are key factors influencing a country's FDI and trade flows by serving as productive inputs in the production (Noorbakhsh, Paloni, and Youssef 2001).

Network factors. Proximity is an important factor affecting FDI and trade flows and we use the broad concept of proximity instead of the narrow spatial proximity in this

Table 2. ERGM results of full sample.

	2015	2013	2011	2009
<i>edges</i>	−2.076***	−1.842***	−2.016***	−2.013***
<i>mutual</i>	2.073***	1.958***	2.064***	1.852***
<i>FDI2015</i>	0.375***			
<i>FDI2013</i>		0.061		
<i>FDI2011</i>			0.042**	
<i>FDI2009</i>				0.098***
<i>POP</i>	0.001***	0.001***	0.001***	0.001***
<i>HC</i>	0.308***	0.315***	0.352***	0.315***
<i>Lang</i>	0.497*	0.551**	0.302	0.217
<i>TA</i>	0.273**	0.271**	0.470***	0.205**

Notes: ***, ** and * denote significance at 1%, 5%, and 10% levels, respectively.

paper. We take social and economic relations as proximity among countries in the sample, such as language, institution, and trade agreement. If two countries have the same language, the value is 1, and otherwise 0. Accordingly, we can construct the language matrix among the countries, represented by *Lang*. *TA* represents the number of trade agreements between two countries.

4.3. Model estimation and results

4.3.1. Results of full sample

The results of full sample in 2009, 2011, 2013, and 2015 are displayed in Table 2. Concretely, the results demonstrate that: (1) edges have significantly negative effect (1% level) on the ITN, which means the ITN among 45 B&R countries is not completely random and supports the characteristics of the ITN in the above section. (2) FDI has the positive effect on trade among 45 B&R countries from the network perspective, which is indicated by the significantly positive coefficients of FDI in 2009, 2011, and 2015 (0.098, 0.042, and 0.375 respectively) and provides support for Hypothesis 1. (3) the positive effect of FDI on trade is improved by the Belt and Road Initiative, which is showed by the significantly increasing of the coefficients from 0.098 and 0.042 in 2009 and 2011, respectively, to 0.375 in 2015, supporting Hypothesis 2.

According to the above results, it confirms that FDI has the positive effect on trade empirically from the multilateral network perspective, in agreement with the researchers who use the Granger causality test to testify the multilateral relations (Dritsaki and Dritsaki 2012; Pourshahabi, Salimi Soderjani, and Mahmoudinia 2013). As to the reason of the positive effect, we can explain it from the following aspects. First, the positive effect is partly because of the enhanced trade relations among 45 B&R countries. The explanation is the increased demand for the products because of the reducing trade costs brought by the Belt and Road Initiative. According to De Soyres et al. (2019), the Belt and Road Initiative significantly reduces shipment times and trade costs by investing in the transport infrastructure, and these effects are magnified by policy reforms of reducing border delays and improving corridor management. The improving efficiency of goods circulation makes the interregional trade more active among the B&R countries and the trade relations become thicker among them. Second, the change of effects of FDI on trade is partly influenced by the core node in the network. In the process of promoting the Belt and Road Initiative, the multilateral investment and trade between China and the B&R countries are escalating. China's direct investment in countries along the B&R totaled

Table 3. Sectoral heterogeneity effects.

	<i>Variables</i>	High	Medium	Low
2015	<i>edges</i>	−2.703***	−2.205***	−2.031***
	<i>mutual</i>	1.006**	1.762***	1.764***
	<i>FDI</i>	0.443**	0.516***	0.507***
	<i>POP</i>	0.001***	0.001***	0.001***
	<i>HC</i>	0.408***	0.315**	0.306*
	<i>Lang</i>	0.216	0.273	0.407*
	<i>TA</i>	0.297***	0.325**	0.216**
2013	<i>edges</i>	−2.604***	−2.317***	−2.425***
	<i>mutual</i>	1.934***	1.706***	1.764**
	<i>FDI</i>	0.093**	0.102**	0.103
	<i>POP</i>	0.001***	0.001***	0.001***
	<i>HC</i>	0.431***	0.364***	0.379***
	<i>Lang</i>	0.314*	0.028	0.031
	<i>TA</i>	0.302**	−0.293*	0.301***
2011	<i>edges</i>	−2.454***	−2.217***	−2.056*
	<i>mutual</i>	1.412*	1.573***	1.063**
	<i>FDI</i>	0.163***	0.102**	0.094**
	<i>POP</i>	0.002***	0.001***	0.001***
	<i>HC</i>	0.317***	0.310***	0.297***
	<i>Lang</i>	0.031	0.406**	0.411**
	<i>TA</i>	−0.042	0.298***	0.271**
2009	<i>edges</i>	−2.647***	−2.296*	−2.052***
	<i>mutual</i>	1.393***	2.057**	1.992***
	<i>FDI</i>	0.203***	0.185***	0.142***
	<i>POP</i>	0.002***	0.001***	0.002***
	<i>HC</i>	0.419***	0.313***	0.289***
	<i>Lang</i>	0.053	0.357	0.322
	<i>TA</i>	0.125	0.270*	0.186

Notes: ***, ** and * denote significance at 1%, 5%, and 10% levels, respectively.

115.63 billion dollars in 2015, which is 180.69% over 2011 and covers the industries of agriculture, manufacturing, and services besides the infrastructure. Meanwhile, China’s trade with the B&R countries has exceeded 6 trillion dollars, and its annual growth rate is higher than that of China’s trade with non B&R countries. Zhang, Tang, and Yi (2019) and Nobi, Lee, and Lee (2020) point out that China acts as a hub along the B&R countries and promotes the industrialized production of the countries with less industrial level by exporting capital and technology.

4.3.2. Sectoral heterogeneity effects

Following Chen, Hsu, and Wang (2012) who suggest that the relationship between FDI and trade varies depending on industry characteristics, we explore how the relationships differ among the sectors having the different technological characteristics further. The heterogeneity results in 2009, 2011, 2013, and 2015 are displayed in Table 3.

First, the coefficients of FDI are positive and significant in three sectors, with the same results for the full sample. This demonstrates that FDI complements trade among the B&R countries regardless of the technological characteristics of the industry. Second, the size of effect of FDI on trade differs among the three sectors, which supports Hypothesis 3. The effect in high technological sector is stronger in 2009 (0.203) and 2011 (0.163) than in the medium technological sector (0.185 and 0.102) and low technological sector (0.142 and 0.094), gauged by the value of the FDI in the two estimations. The results are different in 2013 and 2015, that is, medium technological sector has the

strongest effect (0.102 and 0.516). Also, the coefficient of FDI in the low technological sector (0.507) is bigger than in the high technological sector (0.443) in 2015, which is opposite to the other three years. Apparently, the medium technological sector and low technological sector may be the main beneficiaries of FDI. One explanation is related to the trade structure among the B&R countries. Most of the B&R countries are still in the middle- or low-income stage and some countries still take primary and low technology industries as their pillar industries. Therefore, the increase of intra-regional trade volume is more from the medium technological sector and low technological sector. The other reason is due to the vertical production relations, which is called 'economic complementarity' by Nishitateno (2013). FDI by a final good firm could create demand for parts or intermedium goods, which comes mainly from the medium technological sector.

5. Conclusions and policy implications

Applying the social network analysis methods, this paper examines the effect of FDI on trade using the dataset of B&R countries. Firstly, we describe the characteristics of FDI networks and trade networks. Our findings include: (1) the density and number of edges of the IFN and ITN increase during our study period, which means that the financial and trade relations among 45 B&R countries become more and more closely; (2) both the IFN and ITN are disassortative networks, which represents that central countries preferentially connect to peripheral countries in the network, especially in the ITN; (3) the IFN and ITN are obvious small-world networks and the financial and trade relations are built on the basis of mutually beneficial or cooperative activity; (4) from the nodal view, each country has different importance in the IFN and ITN and can be classified to four groups in the light of out-degree and in-degree centrality.

Second, we explore the effects of FDI on trade by using ERGM further. The empirical estimations indicate that (1) FDI has the positive effect on trade among B&R countries from the network perspective, showing by the results of full sample and sector samples; (2) since the Belt and Road Initiative is proposed, the positive effect of FDI on trade has been improved; (3) FDI promotes the export of products in low technological sector, medium technological sector and high technological sector, but the effect in medium technological sector is stronger than the other two sectors after the Belt and Road Initiative is proposed.

Future research could use the micro enterprise investment data to verify the effect. Some researchers argue that country-level FDI data are not perfect because multinational companies' activities may include investment in real estate which is not related with export (Mitze, Alecke, and Untiedt 2010). Micro FDI flow data may be more appropriate in the FDI network studies. However, it is hard to match the FDI data and trade data in firm level. Despite this, our research gives a comprehensive answer to our research questions that how FDI network impacts the trade network from network evolution perspective.

A key policy implication emanating from this paper is that, FDI can foster trade among B&R countries. Predictably, the investment in infrastructure constructions and the improvement of trade and investment facilitation will further promote the intra-regional trade. However, it still needs international investment rules to regulate direct investment activities. The coming Regional Comprehensive Economic Partnership (RCEP) has the potential to become the supporter for investment agreements among

B&R countries. RCEP contains major countries among B&R countries, such as China, India, and ASEAN. From the perspective of topics, RCEP includes investment promotion, investment protection, and investment facilitation, which can be the main content of the investment agreements among B&R countries.

Notes

1. There are 66 countries in the B&R Initiative. (1) 2 countries in East Asia: China (including mainland, Hong Kong SAR, and Macao SAR) and Mongolia; (2) 10 ASEAN countries: Singapore, Malaysia, Indonesia, Myanmar, Thailand, Laos, Cambodia, Vietnam, Brunei, and Philippines; (3) 18 countries in West Asia: Iran, Iraq, Turkey, Syria, Jordan, Lebanon, Israel, Palestine, Saudi Arabia, Oman, United Arab Emirates, Yemen, Qatar, Kuwait, Bahrain, Greece, Cyprus, and Sinai Peninsula of Egypt; (4) 8 countries in South Asia: India, Pakistan, Bangladesh, Afghanistan, Sri Lanka, Maldives, Nepal, and Bhutan; (5) 5 countries in Central Asia: Kazakhstan, Uzbekistan, Turkmenistan, Tajikistan, and Kyrgyzstan; (6) 7 CIS countries: Russian Federation, Ukraine, Belarus, Georgia, Azerbaijan, Armenia, and Moldova; (7) 16 countries in central and Eastern European: Poland, Lithuania, Estonia, Latvia, Czech Republic, Slovakia, Hungary, Slovenia, Croatia, Bosnia and Herzegovina, Montenegro, Serbia, Albania, Romania, Bulgaria, and TFYR of Macedonia.
2. The 45 economies are (1) 2 countries in East Asia: China (including mainland, Hong Kong SAR and Macao SAR) and Mongolia; (2) 6 ASEAN countries: Singapore, Malaysia, Indonesia, Myanmar, Thailand and Philippines; (3) 7 countries in West Asia: Turkey, Jordan, Lebanon, Israel, Saudi Arabia, Kuwait, Bahrain; (4) 6 countries in South Asia: India, Pakistan, Bangladesh, Sri Lanka, Nepal and Bhutan; (5) 2 countries in Central Asia: Kazakhstan and Tajikistan; (6) 6 CIS countries: Russian Federation, Ukraine, Belarus, Georgia, Azerbaijan and Armenia; (7) 14 countries in central and Eastern European: Poland, Lithuania, Estonia, Latvia, Hungary, Slovenia, Croatia, Bosnia and Herzegovina, Montenegro, Serbia, Albania, Romania, Bulgaria, and TFYR of Macedonia.

Disclosure statement

No potential conflict of interest was reported by the author(s).

Funding

This work was supported by Guangdong Science and Technology Department [grant number 2019A101002064]; Humanities and Social Sciences Research Fund of Ministry of Education of China [grant number 20YJC790189]; The 2020 research project of the 13th Five-Year Plan for Education science of Guangdong Province [grant number 2020GXJK213]; Social Science Planning Project of Guangdong Province [grant number GD19YGL17]; Guangdong Provincial Science and Technology Project [grant number 2019ST008].

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Appendix

Table A1. Classification of industries based on technology.

Classification	Manufactured products
High technology sector	Office/data processing/telecommunications equip, TVs, transistors, turbines, power generating equipment, pharmaceuticals, aerospace, optical/measuring instruments, cameras
Medium technology sector	Passenger vehicles and parts, commercial vehicles, motorcycles and parts, synthetic fibers, chemicals and paints, fertilizers, plastics, iron, pipes/tubes, engines, motors, industrial machinery, pumps, switchgear, ships, watches
Low technology sector	Textile fabrics, clothing, headgear, footwear, leather manufactures, travel goods, pottery, simple metal parts/structures, furniture, jewelry, toys, plastic products

Source: Authors.

Table A2. Top Five countries of FDI in 45 B&R countries (2009 and 2015).

2009			2015		
High-technology sectors	Medium-technology sectors	Low-technology sectors	High-technology sectors	Medium-technology sectors	Low-technology sectors
China, mainland 43.1%	China, mainland 39.3%	China, mainland 54.1%	China, mainland 53.2%	China, mainland 51.4%	China, mainland 72.8%
China, Hong Kong SAR 30.6%	China, Hong Kong SAR 13.1%	China, Hong Kong SAR 8.0%	China, Hong Kong SAR 40.0%	China, Hong Kong SAR 11.7%	Turkey 4.0%
Singapore 17.5%	Singapore 11.8%	Singapore 5.4%	Philippines 3.0%	Singapore 6.5%	China, Hong Kong SAR 3.9%
Philippines 2.3%	India 4.1%	Turkey 4.6%	India 0.8%	India 5.2%	India 3.3%
Hungary 1.5%	Poland 3.8%	India 3.9%	Hungary 0.7%	Turkey 3.2%	Russian Federation 2.2%

Source: Authors.

**Table A3.** Groups of 45 B&R countries based on the degree centrality.

FDI	Group (1)	Bulgaria; Russian Federation; India; Singapore; China
	Group (2)	Poland; Hungary; Slovenia; Turkey; Jordan; Lebanon; Israel; Saudi Arabia; Malaysia; China: Hong Kong SAR; Romania; Georgia; Kuwait; Ukraine
	Group (3)	Lithuania; Croatia; Montenegro; Serbia; TFYR of Macedonia; Belarus; Azerbaijani; Kazakhstan; Pakistan; Nepal; Bangladesh; Sri Lanka; Thailand; Mongolia; Bosnia and Herzegovina; Armenia; Philippines; Indonesia; Bahrain; Latvia Estonia
	Group (4)	Albania; Myanmar; Bhutan; Tajikistan; China: Macao SAR
Trade	Group (1)	China; China: Hong Kong SAR; Poland; Hungary; Slovenia; Croatia; Romania; Bulgaria; Turkey; Singapore; Malaysia; Russian Federation; India
	Group (2)	Serbia; Lithuania; Estonia; Latvia; Israel; Indonesia; Thailand; Philippines; Ukraine; Belarus; Nepal; Bangladesh; Sri Lanka
	Group (3)	Saudi Arabia; Lebanon; Jordan; Kuwait; Georgia; Azerbaijani; Pakistan; Kazakhstan
	Group (4)	Bosnia and Herzegovina; Montenegro; Albania; TFYR of Macedonia; Bahrain; Myanmar; Armenia; Bhutan; Tajikistan; Mongolia; China: Macao SAR

Source: Authors.